

■ Heart Disease ML Project: Complete Presentation & Viva Guide

Step-by-Step Findings, Data Cleaning Insights, Outlier Theory & Examiner Q&A; in Simple English

■ 1. Ready-to-Speak Presentation Script (Deliver in 2-3 Minutes)

"Good morning/afternoon, Sir and everyone.

*Today, I am presenting my project on **Heart Disease Prediction using Machine Learning**.*

The Problem: Cardiovascular diseases are the leading cause of death globally. Early diagnosis can save millions of lives. However, many heart problems develop silently without obvious symptoms. Our goal is to predict if a patient has heart disease using routine clinical tests.

Data Cleaning (Our Core Focus):

We used a dataset of **918 patients**. Initially, standard checks showed zero missing values. But using medical domain knowledge, we discovered a major hidden problem: **172 patients had a Cholesterol value of 0**, and **1 patient had a Blood Pressure of 0**. In real human biology, a living person cannot have 0 BP or 0 cholesterol! We exposed these masked zeros as NaNs, visualized them with a missing data heatmap, and cleanly imputed them using feature means. We also used the **IQR method** to detect and analyze clinical outliers.

Machine Learning & Results:

We tested multiple classification algorithms. **Logistic Regression** achieved the best balance of performance with **89.13% Accuracy**, a **0.9029 F1-score**, and high **Recall (91.18%)**. In healthcare, Recall is critical because we cannot afford to miss a sick patient.

Web Deployment:

We saved the trained model, scaler, and column structure using joblib and deployed an interactive **Streamlit Web Application** where doctors can enter patient numbers, view an instant risk percentage gauge, and download diagnostic reports.

Thank you. I am now ready for questions and a live demo!"

■ 2. Step-by-Step Walkthrough: What We Did, Found & Concluded

Step & Action	What We Found (Discovery)	What We Concluded (Conclusion / Takeaway)
Step 1: Load Data pd.read_csv	Dataset has 918 rows, 11 features, and 1 target (HeartDisease).	Sufficient sample size to train standard machine learning classifiers.
Step 2: Initial Check isnull().sum()	Output showed 0 null values and 0 duplicate rows.	Superficially looks clean, but requires deeper domain checks for invalid values.
Step 3: Visual EDA Target & Feature plots	- Target: 55.3% disease (508), 44.7% normal (410). - Males have higher disease rate. - Asymptomatic chest pain has highest risk.	Target is well-balanced (no severe class imbalance). Certain symptoms strongly correlate with disease.
Step 4.1: Expose Zeros Domain Check	172 rows had Cholesterol = 0 (18.7% of data!). 1 row had RestingBP = 0.	These are hidden missing values entered as 0. Must be converted to NaN and cleaned.
Step 4.2: Imputation Mean Replacement	Replaced NaNs with positive means: 244.64 mg/dl for Cholesterol, 132.54 mm Hg for BP.	Before/After plots prove the fake 0-spike disappeared and formed a clean bell curve.
Step 4.3: Outliers IQR Boxplot Analysis	Found 28 BP outliers (>170), 183 Cholesterol outliers, 16 Oldpeak outliers (>3.8).	Outliers reflect severe medical conditions (hypertension/hypercholesterolemia), not errors.
Step 5: Encoding & Split get_dummies & split	Transformed categorical text into 15 numeric columns. Split 80% train (734), 20% test (184).	Stratification ensures train and test sets have exact same 55% disease proportion.
Step 6: Scaling StandardScaler	Scaled mean=0, std=1 on X_train. Applied transform on X_test.	Prevents high-range features (Cholesterol 250) from dominating small features (Oldpeak 1.5).
Step 7: Comparison 6 Classifiers Tested	Logistic Regression (89.1%), Random Forest (88.6%), SVM (88.0%), KNN (85.8%).	Logistic Regression is best: high accuracy, high recall, and clear explainability.
Step 8: Detailed Eval Confusion Matrix	True Positives: 93, True Negatives: 71, False Negatives: 9, False Positives: 11.	Very high Recall (91.18%) ensures only 9 out of 102 diseased patients were missed.
Step 9: Feature Impact Coefficients Bar Plot	Flat/Down ST_Slope, Exercise Angina, and Oldpeak have largest positive weights.	Confirms clinical truth: stress-induced ECG abnormalities are top predictors of heart disease.
Step 10: Artifact Export joblib.dump	Saved model (.pkl), scaler (.pkl), and feature column names list (.pkl).	Provides complete standalone backend to power the Streamlit web interface.

■ 3. In-Depth Concepts: Outliers & Data Cleaning (Examiner Favorites)

A. What are Outliers and How Do We Handle Them?

Definition: An outlier is a data point that is extremely distant from the rest of the observations in the dataset.

Why do they happen? (1) Measurement or recording errors, or (2) Natural clinical extremes (e.g., a patient in acute hypertensive crisis with BP = 200 mm Hg).

How did we detect them? We used the **Interquartile Range (IQR) Method**:

- **Q1 (25th percentile):** Middle value of the first half.
- **Q3 (75th percentile):** Middle value of the second half.
- **IQR = Q3 - Q1**
- **Lower Bound = Q1 - 1.5 × IQR | Upper Bound = Q3 + 1.5 × IQR**

Any point beyond these bounds is an outlier.

How do we handle them in real projects?

1. *Trimming (Removal):* Dropping rows. (Bad for small medical datasets; loses real sick patients).
2. *Capping (Winsorization):* Replacing values above the upper bound with the upper bound value.
3. *Keeping with Robust Scaling (Our choice):* In medicine, an extreme BP of 200 is real clinical signal, not noise. Standardizing preserves this signal.

B. The Masked Zeros Problem (Domain-Specific Cleaning)

The Discovery: A standard `df.isnull().sum()` showed 0 nulls. But medical logic dictates:

- A human heart stopped pumping if **Blood Pressure = 0 mm Hg**.
- A living human cell cannot exist with **Total Cholesterol = 0 mg/dl**.

Therefore, 172 rows had Cholesterol = 0 because doctors didn't record it! We converted them to `np.nan` to expose the missing values and imputed them using the mean of non-zero entries (244.64 mg/dl).

C. Feature Scaling & Data Leakage Prevention

Why Scale? Age is 20-80, BP is 80-200, Cholesterol is 100-500, while Oldpeak is 0-4. Without scaling, gradient-based models treat larger numbers as more important.

What is Data Leakage? If you fit the scaler on the whole dataset before splitting, the mean and standard deviation of the test set leak into training! To prevent this: `scaler.fit_transform(X_train)` on training data only, and `scaler.transform(X_test)` on test data.

D. Confusion Matrix & Healthcare Evaluation Metrics

In disease prediction, standard accuracy is not enough. We look at the Confusion Matrix:

- **True Positive (TP = 93):** Sick patient correctly diagnosed with heart disease.
- **True Negative (TN = 71):** Healthy person correctly diagnosed as normal.
- **False Positive (FP = 11):** Healthy person falsely told they have heart disease (Causes extra tests, but not fatal).
- **False Negative (FN = 9): DANGEROUS!** Sick patient sent home being told they are healthy.

Recall (Sensitivity) = $TP / (TP + FN) = 93 / (93 + 9) = 91.18\%$. Our model achieves over 91% recall, minimizing fatal false negatives!

■ 4. Top 12 Viva & Technical Interview Questions with Answers

Q1. What is the objective of your project?

Ans: To build an end-to-end machine learning system that predicts heart disease risk using clinical patient biomarkers, with a focus on domain data cleaning and an interactive web frontend.

Q2. Why did you impute zeros in Cholesterol and Blood Pressure instead of dropping them?

Ans: Dropping 172 rows would delete 18.7% of our dataset, reducing training diversity and statistical power. Mean imputation preserves those patients while filling the missing value with a clinically reasonable average (244.64 mg/dl).

Q3. What is an outlier and what formula did you use to detect them?

Ans: An outlier is an extreme value distant from the rest. We used the Interquartile Range (IQR) method: Lower = $Q1 - 1.5 \times IQR$, Upper = $Q3 + 1.5 \times IQR$, where $IQR = Q3 - Q1$.

Q4. Why did you keep outliers instead of deleting them?

Ans: In medicine, extreme values like BP = 200 or Cholesterol = 450 are not measuring errors; they represent genuine acute patient crises that strongly predict heart disease. Deleting them would remove the exact high-risk cases we want to catch.

Q5. Why did you use one-hot encoding with drop_first=True?

Ans: Categorical features like ChestPainType (ASY, NAP, ATA, TA) have no numeric order. One-hot encoding creates binary columns. Setting drop_first=True eliminates one category to avoid the dummy variable trap (multicollinearity).

Q6. Why did you use StandardScaler?

Ans: Features have vastly different scales (Cholesterol ~250 vs Oldpeak ~1.5). StandardScaler centers data to mean=0 and variance=1 so the optimizer converges efficiently and treats all features fairly.

Q7. What is data leakage and how did you avoid it?

Ans: Data leakage happens when test set information leaks into training. We avoided it by splitting the dataset into 80:20 BEFORE scaling, fitting the scaler strictly on X_train, and only transforming X_test.

Q8. Why choose Logistic Regression over complex models?

Ans: Logistic Regression achieved our highest accuracy (89.13%) and high recall (91.18%). More importantly, healthcare requires explainability — doctors need to see feature weights and calibrated probabilities, not an opaque black box.

Q9. Why is Recall more important than Accuracy in this project?

Ans: Accuracy treats all mistakes equally. In medicine, a False Negative (missing a sick patient) can lead to untreated death, whereas a False Positive only causes a harmless follow-up check. High Recall (91.18%) minimizes false negatives.

Q10. What are the top features that cause Heart Disease in your model?

Ans: ST_Slope (Flat/Down), Exercise-Induced Angina (Yes), and Oldpeak ST Depression (>1.5 mm). These are all direct indicators that the heart is starved of oxygen during physical exertion.

Q11. Why do we export .pkl files using joblib?

Ans: To decouple model training from deployment. The web application loads the pre-trained model and scaler instantly in 0.1 seconds without needing to retrain from scratch every time a user visits.

Q12. What are the limitations or future scope of this project?

Ans: The dataset has 918 records; collecting larger multi-center clinical trials would improve generalization. In future work, we can integrate SHAP values for personalized explainability and connect with hospital Electronic Health Records (EHR).